

Jiajie Wu CV

Contact Information

Email: j.wu24@imperial.ac.uk

Education

PhD in Mathematics, Imperial College London

From 2024.09

M.S. in Mathematics, New York University

2022.09 - 2024.05

B.A. in Mathematics, New York University (with high honors)

2018.09 - 2022.05

Research Interests

Active Matter, Bio-fluid Dynamics, Cilia Dynamics, Collective Locomotion in Fluids, Delay Systems, Nonlinear Dynamics

Research Projects

Research at the NYU Applied Math Lab

2019.11-2024.06

Fluid-structure Interactions within Bird Flocks in Linear Formation Flights

Hydrodynamics of Finite-length Pipes at Intermediate Re

The Water Clocks

Research at Imperial College London

Since 2024.10

The Collective Dynamics of Cilia

Research at Helmholtz Pioneering Campus (Imperial-TUM Partnership)

Since 2024.11

The Physics of Mucus

Publications

C. Mavroyiakoumou, **J. Wu**, and L. Ristroph. "Modeling flying formations as flow-mediated matter," *Phys. Rev. Fluids* 11, 063103 (2026).

J. W. Newbolt, N. Lewis, M. Bleu, **J. Wu**, C. Mavroyiakoumou, S. Ramanarivo, and L. Ristroph. "Flow interactions lead to self-organized flight formations disrupted by self-amplifying waves," *Nat. Commun.* 15, 3462 (2024).

O. Pomerenk, S. Carrillo Segura, F. Cao, **J. Wu**, and L. Ristroph. "Hydrodynamics of finite-length pipes at Intermediate Reynolds Numbers," *J. of Fluid Mech.* (2023), vol. 959, A28.

Awards and Honors

Imperial-TUM Joint Academy of Doctoral Studies Scholarship (MAAP_G44167)

2024-2027

Roth Scholarship (MATH_G00700)

2024-2028

Graduate School of Arts & Science Tuition Scholarship

2022-2023

Phi Beta Kappa Honors Society

2022.04

Founders Day Award (University Honors Scholar)

2022.04

Larry & Klara Silverstein Research Scholar

2020.05

Dean's Undergraduate Research Fund

2020.05

Dean's List for Academic Year (all 4 academic years, 2018 - 2022)

Research Talks

The Physics of Mucus, Joint Academy of Doctoral Studies Summer School

2026.07

The Water Clocks, Dean's Undergraduate Research Conference

2021.05

Work Experience

Math & English Tutor, Shanghai Qingtu Education

Summer 2019

Course Grader, NYU Math Department

MA-UY 1124 Calculus II for Engineers

Fall 2020

MA-UY 4414 Applied Partial Differential Equations

Fall 2020

Graduate Employee Adjunct Faculty, Courant Institute of Mathematical Sciences

MATH-UA 9 Algebra, Trigonometry & Functions	Fall 2022
MATH-UA 264 Chaos and Dynamical Systems	Spring 2023
MATH-UA 132 Math for Economics II	Fall 2023

Graduate Teaching Assistant, Imperial College London

MATH50004 Multivariable Calculus and Differential Equations	Fall 2024
MATH40007 Introduction to Applied Mathematics	Spring 2025
MATH50003 Linear Algebra and Numerical Analysis	Spring 2025
MATH50003 Linear Algebra and Numerical Analysis	Fall 2025
MATH50003 Linear Algebra and Numerical Analysis	Spring 2026

References

Prof. Eric Keaveny, Professor of Applied Mathematics, Imperial College London, e.keaveny@imperial.ac.uk

Prof. Leif Ristroph, Associate Professor of Mathematics, New York University, lr1090@nyu.edu

Prof. Michael Shelley, Lilian and George Lyttle Professor of Applied Mathematics, New York University and Director of Center of Computational Biology, Flatiron Institute, mshelley@flatironinstitute.org